



Database .sqlite .duckdb

Table: file_info			
filename	n_scans	timestamp	Etc.
Source file	Scan count	Timestamp	...
Smp_A	1445	2021-08-23 04:30	...
Smp_B	1489	2021-08-23 05:25	...
...additional samples...			
Smp_Y	1388	2021-08-23 22:15	...
Smp_Z	1390	2021-08-23 23:10	...

Table: scan_info					
filename	scan_num	ms_level	rt	TIC	Etc.
Source file	Scan number	MS level	Retention time	Ion current	...
Smp_A	1	1	0.10	2331809	...
Smp_A	2	2	0.12	820427	...
...many additional scans...					
Smp_Z	1389	2	22.43	9555703	...
Smp_Z	1390	2	22.45	1892338	...

Table: MS1				
filename	scan_num	rt	mz	int
Source file	Scan number	Retention time	m/z ratio	Intensity
Smp_A	1	0.10	60.0452	6618
Smp_A	1	0.10	60.0532	2657
...millions of additional entries...				
Smp_Z	1385	22.35	60.0456	158084
Smp_Z	1385	22.35	60.0531	4673

Table: MS2						
filename	scan_num	prescan	rt	fragmz	premz	int
Source file	Scan number	Precursor scan	Retention time	Fragment m/z	Precursor m/z	Intensity
Smp_A	2	1	0.12	51.0238	241.0894	36104
Smp_A	2	1	0.12	53.0394	241.0894	243165
...millions of additional entries...						
Smp_Z	1390	1385	22.45	52.0186	185.1932	28371
Smp_Z	1390	1385	22.45	57.0923	185.1932	129604

Ion chromatogram extraction: `SELECT * FROM MS1 WHERE mz BETWEEN min AND max`

Retention time range subset: `SELECT * FROM MS1 WHERE rt BETWEEN min AND max`

Fragment search: `SELECT * FROM MS2 WHERE fragmz BETWEEN min AND max`

Precursor search: `SELECT * FROM MS2 WHERE premz BETWEEN min AND max`